04-Quad-exc

Numeric Quadrature

1. Use the composite Simpson's formula on five points $x_k = kh$, $0 \le k \le 4$, $h = 0.25$, 207b2d

   (a) to approximate $\int_0^1 f(x)\,dx$, where $f(x) = \begin{cases} (e^x - 1)/x, & x \neq 0, \\ 1 \quad , & x = 0. \end{cases}$

   (b) But how accurate is what you got? Theory says that composite Simpson quadrature $Q_h F$ for smooth functions at step h leaves a truncation error

   $$|Q_h - \int_a^b f(x)\,dx| = \frac{f^{(4)}(\xi)}{180}(b-a)h^4, \quad \text{some} \quad a < \xi < b,$$

   Hopefully $f^{(4)}(\xi) \sim f^{(4)}(0)$, and it is easy ($e^x = 1 + x + x^2/2! + \cdots$) to rip off the value of $f^{(4)}(0)$ from the Maclaurin expansion of $f(x)$. Do it and estimate the order of your quadrature error!

2. Devise two routines, `qTrap(yi,h)` and `qSimp(yi,h)` which realize the composite Trapez and Simpson rules for numerical quadrature. Input parameters: `yi` - a list of function values computed on a regular grid of step size `h`. For the composite Simpson rule the # of nodes `len(yi)` should be odd. af30c1

```
import numpy as np  # Example Python code:

def qTrap(yi,h):    # Trapez rule quadrature
    T  = 0.5*(yi[0]+yi[-1]) # Endnodes
    T += np.sum(yi[1:-1])   # Innernodes
    return T*h

def qSimp(yi,h):    # Simpson 1/3 rule
    n = len(yi)
    if (n < 3) or (n % 2 == 0):
        print(f"ERROR: #(nodes) should be >=3 and ODD")
        return None
    S  = yi[0] + yi[-1]          # Endnodes
    S += 2.0*np.sum(yi[2:-2:2])  # Interval endnodes
    S += 4.0*np.sum(yi[1:-1:2])  # Interval midpoints
    return S*h/3
```

   (a) Check your routines by integrating, e.g., $\int_0^{\pi/2} \cos(x)\,dx = 1$, at stepsizes $h_k = \frac{\pi}{8k}$, $k = 1, 2, 3, 4$.

   (b) Find the approx. order: If the error follows $\epsilon_k = C h_k^\alpha$, then the regression line through $\{\log h_k, \log \epsilon_k\}$,

   $$\boxed{\log \epsilon = \log C + \alpha \log h}, \quad \text{least squares solution of} \quad \begin{bmatrix} 1 & \log h_1 \\ 1 & \log h_2 \\ 1 & \log h_3 \\ 1 & \log h_4 \end{bmatrix} \begin{bmatrix} \log C \\ \alpha \end{bmatrix} = \begin{bmatrix} \log \epsilon_1 \\ \log \epsilon_2 \\ \log \epsilon_3 \\ \log \epsilon_4 \end{bmatrix},$$

   has a gradient of α and can be retrieved, e.g., by using Python's `np.polyfit(logH,LogEps,1)`.

3. Compute the error $\epsilon_h = I - Q_h$ of the numerical quadrature of $I = \int_0^{\pi/6} \cos x\,dx$ using the Trapez method 15040d
   at step lengths $h = \frac{\pi}{6}$ and $\frac{\pi}{12}$, then compare with the theoretical truncation error $\epsilon_h = \frac{|f''(\xi)|}{12} h^2 (b-a)$.

   Interpolate by Romberg's method between Q_h and $Q_{\frac{h}{2}}$ to boost convergence to $I - \mathcal{R}Q_h = O(h^4)$. State explicitly the formula you use, compute $\mathcal{R}Q_h$ using 6 significant digits and give the error.

4. The Simpson composite rule delivers $\mathcal{O}(h^4)$-order of convergence only for functions which are C^4 or smoother on the closed interval of integration. Motivate which one of the following rewritings of the weak singular integral to the right is best with regard to numerical quadrature: cba609

$$I = \int_0^1 \frac{\sqrt{4-x^2}}{\sqrt{1-x^2}}\,dx$$

(a) change variables $t = \arcsin(x)$ (b) change variables $t = \sqrt{1-x}$;

(c) partial integrate once, using that $\arcsin'(x) = (1-x^2)^{-1/2}$.

Optional: Use your routines `qTrap` and `qSimp` at two step sizes h_1 and h_2, say, by subdividing the interval in $n_1 = 4$ and $n_2 = 8$ subintervals and estimate the order of convergence, by

$$\alpha = \frac{\ln\epsilon_2 - \ln\epsilon_1}{\ln h_2 - \ln h_1} = \frac{\ln(\epsilon_1/\epsilon_2)}{\ln 2} \simeq 1.44 \ln\frac{\epsilon_1}{\epsilon_2}, \quad \epsilon_i = |I - Q_{h_i}^{(2)} f|, \quad I = 2.93492441867885431\ldots.$$

5. You plan to compute I with error 10^{-10}. Assume you are done if you reduce the job to numerical quadrature of a smooth function over a finite interval, as it can be carried out effectively to any given precision.

$$I = \int_0^\infty \frac{1-e^{-x^3}}{x^{1.3}}\,dx$$

(a) Split the integration $I_0^\infty = I_0^1 + I_1^a + I_a^\infty$, where $a > 1$ is to be fixed later.

(b) In I_0^1: Expand e^{-x^3} in series, integrate, and truncate the alternating series at the 10^{-10}-term.

(c) I_1^a has a C^∞ integrand, ready for effective numerical quadrature for any finite a.

(d) I_a^∞: Make an estimate to choose a so that $I_a^\infty < 10^{-10}$, so it can be discard it.

Optional: In (b), use your routines `qTrap` and `qSimp` to subdivide the interval in 10, 20, 40 and 80 to evaluate the integral. How does Romberg integration perform using the data from `qTrap`? Compare:

$$I_0^1 \simeq 0.298743965125389 \qquad I_1^3 \simeq 0.8659413975189196 \qquad I_3^\infty \simeq 2.397410311082881.$$

6. Simplify the integral $I = \displaystyle\int_0^1 \frac{e^{0.04t^2}-1}{t}\,dt$ by changing variables $x = 0.2t$. Expand the integrand in Taylor series, integrate term-wise and truncate to find an approximation of I with five significant digits. 5215de

7. The two integrals I_1 and I_2 are approximated at two different step lengths $h_{1,2}$ using the composite Trapez and Simpson rules with quadrature errors $\epsilon_i = |Q(h_i) - I|$ in the tables below: 3cef9a

$$I_1 = \int_0^1 \frac{\ln(1+x)}{x}\,dx = \frac{\pi^2}{12} \rightsquigarrow$$

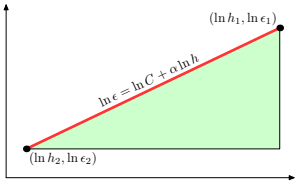
h_i	Trapetz	Error ϵ_i	Simpson	Error ϵ_i
1.250e-01	0.822866	4.0e-04	0.822468	1.8e-06
3.125e-02	0.822492	2.5e-05	0.822467	7.1e-09

$$I_2 = \int_0^1 (1+\arcsin x)\,dx = \frac{\pi}{2} \rightsquigarrow$$

h_i	Trapetz	Error ϵ_i	Simpson	Error ϵ_i
1.250e-01	1.582503	1.2e-02	1.575860	5.1e-03
3.125e-02	1.572339	1.5e-03	1.571430	6.3e-04

For smooth functions, the leading error term in ϵ is of order α, $\epsilon = Ch^\alpha$, with $\alpha = 2$, $C = C(f'')$, for the composite Trapetz rule, and $\alpha = 4$, $C = C(f^{(4)})$, for the Simpson rule.

(a) In log scales $\ln\epsilon = \ln C + \alpha \ln h$. Interpolate in the tables above to obtain estimates for the rate α in $I_{1,2}$ for both quadrature rules.



(b) Why the approximation rates α for the seemingly singular I_1 do match the ones for smooth functions while the rates for I_2 are worse?

Hint: $\ln(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \dfrac{x^4}{4} + \cdots$, $|x| < 1$.

Rewrite as Special functions

1. Change variables to express the weak singular integral I to the right through the standard special function (the complete elliptic integral of the second kind) cba610

$$I = \int_0^1 \frac{\sqrt{4-x^2}}{\sqrt{1-x^2}}\,dx$$

$$E(k) = \int_0^{\pi/2} \sqrt{1-k^2\sin^2 t}\;dt.$$

2. Split the integral to separate the singularities and tackle resulting integrals to become suitable for numerical quadrature. (Or, complete squares, shift by one, and nail the integral analytically.) 29d1d6

$$I = \int_0^2 \frac{dx}{\sqrt{2x-x^2}}$$

3. The well-known (and readily available from libraries of special functions) Euler Gamma function, d4a5a6

$$\Gamma(z) = \int_0^\infty t^{z-1}\,e^{-t}\,dt,$$

is well-known generalization of the factorial for any $z \in \mathbb{C}$ with $\operatorname{Re} z > 0$.

   (a) Show by induction that $\Gamma(n+1) = n!$ for any integer $n \geq 0$.

   (b) Express the integral $I = \int_0^\infty x^e\,e^{-\pi x}\,dx$ in terms of the Gamma function.

   Optional: Compute I using some implementation (Matlab/Octave, Python, Julia,…) of $\Gamma(z)$.

4. Change variables $x = e^{-t}$ in the integral 7be815

$$I = \int_0^1 \frac{dx}{\sqrt{-\ln x}}$$

in order to express it in terms of the well-known Euler Gamma function:

$$\Gamma(z) = \int_0^\infty t^{z-1}\,e^{-t}\,dt.$$

5. Split the integral and then make a suitable change of variables to express the probability 2e8b92

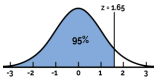
$$\text{Prob}(X \leq \sqrt{2}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\sqrt{2}} e^{-x^2/2}\,dx$$

of a standard normal random variable X through the Gauss error function

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2}\,dt. \qquad \text{(Note that Prob}(X \leq 0) = 0.5\text{)}.$$

Parallelized

1. The cumulative probability function of the standard normal distribution is given by

$$Z \sim N(0,1): \quad \text{  } \quad \rightsquigarrow \quad \text{Prob}(Z \leq z) = \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-t^2/2}\,dt.$$

The error function is readily available in libraries of special mathematical functions, and defined by 253b8f

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,dt.$$

For example:

x	0	0.1	0.2	0.3	0.4	0.5	$\cdots$
erf(x)	0	0.1125	0.2227	0.3286	0.4284	0.5205	$\cdots$

   (a) Change variables to express $\Phi(z)$ through the error function erf. (Of course, $\Phi(0) = \frac{1}{2}$.)

   (b) Use (a) and then interpolate in the table above to find $\Phi(0.324)$.

2. (Rough assessment of convergence rate α) Suppose that the absolute value of the error ϵ of a finite difference scheme approximating a certain quantity follows the power low to the right as a function of the step size h. Show that having error counts $\epsilon_i(h_1)$ and $\epsilon_{i+1}(h_{i+1})$ at two resolutions, then α can be ripped,

$$\left\{\begin{matrix} h_i \\ h_{i+1} \end{matrix}\middle|\begin{matrix} \epsilon_i \\ \epsilon_{i+1} \end{matrix}\right. \Rightarrow \left\{\begin{matrix} \ln h_i \\ \ln h_{i+1} \end{matrix}\middle|\begin{matrix} \ln \epsilon_i \\ \ln \epsilon_{i+1} \end{matrix}\right. \qquad \alpha = [\ln \epsilon_i : \ln \epsilon_{i+1}] \qquad \alpha = \frac{\ln(\epsilon_i/\epsilon_{i+1})}{\ln(h_i/h_{i+1})},$$

as the senior coefficient of the linear Newton IP in logarithmic coordinates. Fill in the table!

$$\boxed{\epsilon(h) = Ch^\alpha}$$

h_i	ϵ_i	α
0.500	4.32 e-3	
0.250	1.41 e-3	?
0.125	4.76 e-4	?